Journal of Informetrics 16 (2022) 101311

Contents lists available at ScienceDirect

Journal of Informetrics

journal homepage: www.elsevier.com/locate/joi

# Movers' advantages: The effect of mobility on scientists' productivity and collaboration

Meijun Liu[a,b,c], Xiao Hu[d,e,*]

[a] Institute for Global Public Policy, Fudan University, Shanghai, China
[b] LSE-Fudan Research Centre for Global Public Policy, London, UK
[c] Shanghai Center for Innovation and Governance, Fudan University, Shanghai, China
[d] Faculty of Education, University of Hong Kong, Hong Kong, China
[e] Musketeers Foundation Institute of Data Science, University of Hong Kong, Hong Kong, China

ARTICLE INFO

*Keywords:*
Scientific mobility
Scientific collaboration
Productivity
Difference in differences
Propensity score matching

ABSTRACT

With the rapid globalization of science, mobility is perceived as an important driver of scientific progress and innovation success. However, we have little knowledge about whether and how scientists' mobility influences their career development, especially scientists' productivity and collaboration. In this case study, using the data on 62,330 scientists, the Chinese computer scientists who published at least one computer science paper and published no fewer than 10 papers in total from 2000 to 2012, we apply difference in differences models in conjunction with PSM methods to show the effect of domestic mobility (i.e., moving inside China) on scientists' research quantity and quality by distinguishing the direction of mobility. In contrast to the existing literature that documents a short-term negative effect due to adaption costs or disruption of routines and social capital, we do not observe an initial detrimental impact of following moves on productivity and collaboration, even for non-upward moves. We further find that mobility leads to increased collaboration with new partners without dampening scientists' collaboration with previous collaborators. However, scientists have a higher probability of collaborating with new collaborators, as evidenced by the decreased share of previous collaborators to the total co-authors after they move. The findings of this case study imply that the benefits of mobility might outweigh its costs and that mobility improves scientists' productivity and collaboration for prolific scientists in emerging countries.

## 1. Introduction

Circulation of the scientific workforce has been a global phenomenon and is considered important to the production of knowledge. The mobility of scientists across countries, fields, sectors and institutions is increasingly encouraged at the policy level in many countries to improve the performance of the research system (Fernandez-Zubieta, Geuna, & Lawson, 2013; Jones, 2021; Scellato, Franzoni, & Stephan, 2015; Woods, 2022). According to the knowledge recombination theory, innovative discoveries and creative ideas are often generated from distant sources (Fleming, 2001; Hargadon & Sutton, 1997), which is crucial for innovation success. With knowledge embedded in moved scientific labor, mobility benefits the research system by facilitating knowledge exchange between different universities and regions (Azoulay, Zivin, & Sampat, 2011; Sugimoto et al., 2017).

Although mobility is generally considered beneficial to the research system, there are ongoing debates on the effect of mobility on individuals' career advancement. On the one hand, mobility is considered crucial for developing scientists' career and improving their

* Corresponding author.
*E-mail address:* xiaoxhu@hku.hk (X. Hu).

https://doi.org/10.1016/j.joi.2022.101311
Received 9 October 2021; Received in revised form 14 June 2022; Accepted 16 June 2022
Available online 30 June 2022
1751-1577/© 2022 Elsevier Ltd. All rights reserved.

research performance since it offers diverse career opportunities (Fortunato et al., 2018), improves match quality between scientists and their employers (Hoisl, 2007; Topel & Ward, 1992), and increases scientists' human and social capital (Fernández-Zubieta, Geuna, & Lawson, 2015). However, significant costs that dampen scientists' productivity and collaboration could be generated when scientists change affiliations because of professional and personal dislocation (Azoulay, Ganguli, & Zivin, 2017), adjustment costs (Fernández-Zubieta, Geuna, & Lawson, 2016), disruption of routines and social connections (Azoulay et al., 2017; Groysberg, Lee, & Nanda, 2008).

Despite the importance of mobility for scientific productivity, reward and hiring policies, we have a limited quantitative understanding of how such mobility shapes individuals' careers, especially scientists' productivity and collaboration. Another challenge is the endogeneity problem caused by a positive selection of moved scientists. Besides, science is characterized by a high degree of stratification as resources are highly concentrated in a few individuals or institutions (Davies & Zarifa, 2012; Merton, 1968; Salmi, 2010). The investigation of the effect of mobility on scientists' productivity and collaboration should take the direction of mobility (e.g., from institutions with little resources to those with rich resources) into account.

Performing a case study of more than 62,000 productive Chinese scientists working in the field of computer science (i.e., the Chinese scientists who published at least one computer science research article and published no fewer than 10 papers in total) from universities, this paper fills the gap by examining the influence of mobility within the country on scientists' productivity and collaboration using the PSM-DID estimators to deal with the endogeneity problem. We differentiate the direction of mobility and reveal the dynamic effect of scientists' mobility.

The mobility of productive computer scientists is the research focus for several reasons. These scientists significantly contribute to pushing the boundaries of the knowledge frontier in their field. Changes in work environment that enhance their scientific human and social capital can increase social returns by promoting information and technology innovation, and generating positive social externalities. Second, team efforts and scientific collaboration are important for knowledge production in computer science. Gains or losses of scientific social capital that are relevant to mobility can be manifested in their publishing activities. Last, according to the power-law distribution of scientific items (Brzezinski, 2015; Lehmann, Lautrup, & Jackson, 2003), such as publications and citations, a small proportion of scientists account for a majority of knowledge production and innovation.

This study contributes to existing literature by focusing on the research contexts of emerging economies and within-country mobility, which are less investigated. Scientists' mobility has been extensively discussed for some scientific powerhouses in the West, e.g., the U.S. and European countries (Fernández-Zubieta, Geuna, & Lawson, 2015; Hoisl, 2007; Tartari, Di Lorenzo, & Campbell, 2018), while the investigation of scientists' mobility across institutions in emerging economies is scarce. Considering the huge differences in research systems, science policies and cultures between emerging economies and advanced ones (Wang, Hu, & Liu, 2017), the existing findings about scientific mobility in the latter context may not be well applied to the former. The exploration of Chinese scientists' mobility could be a valuable case for understanding scientific mobility in emerging economies. Besides, this study contributes to the existing literature on mobility by providing a one-country perspective. We focus on domestic mobility (i.e., mobility across domestic affiliations within the country) which is less explored previously, rather than international mobility (i.e., mobility between a foreign affiliation and a Chinese affiliation) that most prior studies investigated (Cañibano, D'Este, Otamendi, & Woolley, 2020; Fangmeng, 2016; Li & Tang, 2019; Netz, Hampel, & Aman, 2020).

The remainder of the paper is organized as follows. The next section reviews the related work and proposes the research questions. Section 3 lays out the details of data and our empirical approaches. In section 4, the results are presented. The last section presents a discussion of the findings and implications for relevant policies.

## 2. Literature review and research questions

### 2.1. Mobility and productivity

Increased productivity is expected after mobility because of better match quality of employers and employees, knowledge spillovers from new colleagues and better access to resources. The match quality between employees and employers could be improved by mobility (Topel & Ward, 1992). For example, looking at both domestic mobility and international mobility, Hoisl (2007) attributes a positive effect of mobility on inventors' productivity to an improvement of the match between employers and employees, and knowledge flow from movers' new colleagues. Previous research provides evidence on knowledge spillover and peer effects embedded in scientists' international mobility (Borjas & Doran, 2012; Franzoni, Scellato, & Stephan, 2014; Moser, Voena, & Waldinger, 2014; Scellato, Franzoni, & Stephan, 2012; Waldinger, 2012) or general mobility that includes both domestic and international mobility (Agrawal, Cockburn, & McHale, 2006). A majority of current studies provide descriptive evidence to show that mobile scientists are more productive and cited than their static counterparts by exploring international mobility (Franzoni et al., 2014), general mobility (Aksnes, Rørstad, Piro, & Sivertsen, 2013; De Filippo, Casado, & Gómez, 2009; Manuel, Shiff, & Melamed, 2006), and mobility within the country (Jaffe, Trajtenberg, & Henderson, 1993). Using more sophisticated econometric techniques, such as instrumental variable approaches, some researchers address the causality issue between general mobility and increased productivity (Hoisl, 2007; Tartari, Di Lorenzo, & Campbell, 2018).

In contrast, no effect or negative impacts of mobility on productivity are found in other studies. Mobility generates costs including direct personal costs and skill-adjustment costs, especially for high-skilled positions (Fernández-Zubieta et al., 2016). Scientists who move to a new affiliation have to adjust themselves to the new environment by learning new practices, protocols, and routines. Van Heeringen and Dijkwel (1987) find no effect of job mobility without changing the research field on research output. Performing a Curriculum Vitae-based analysis, Cañibano, Otamendi, and Andújar (2008) focus on international mobility and pro-

vide empirical evidence to support no effect of international mobility on research quantity. Exploring general mobility, Fernández-Zubieta et al. (2016) document a short-term decline in productivity after mobility due to adjustment costs. Groysberg et al. (2008) find that star security analysts experience an immediate reduction in performance after changing employers and attribute this negative effect to the disruptions of routines and social capital. A more recent study presents that domestic mobility has negative impacts on researchers' research output at least in the short term based on data on economics and management researchers, and this negative effect might be driven by researchers' loss of social capital from the doctorate-granting institution (Bäker, 2015).

The current literature presents several research gaps. It is unclear whether job mobility improves or dampens scientists' productivity given the conflicting results obtained in prior studies. The investigation of the effect of mobility on individual productivity is mainly characterized by the challenge caused by the endogeneity issue. This is because productive scientists are inclined to move to a new affiliation as they have more mobility options considering organizations' deliberate choices of job candidates (Azoulay et al., 2017; Gibson & McKenzie, 2012; Grogger & Hanson, 2011; Singh & Agrawal, 2011; Van Heeringen & Dijkwel, 1987).

### 2.2. Mobility and collaboration

Scientists' prior connections constructed in their previous affiliations might not be portable after they move. Collaboration involves expenditure in time, resources and intensive communication and coordination of activities, which tends to be started and sustained through interactions that happen in person (Wagner, 2008). Due to the interpersonal nature (Groysberg et al., 2008; Katz, 1994) and the importance of physical proximity (Jaffe, Trajtenberg, & Henderson, 1993) for scientific collaboration, it is argued that scientists' collaboration and other social capital developed in their former employers may be dampened if they change to a new affiliation (Groysberg et al., 2008).

However, another strand of literature provides evidence on scientists' enduring and transportable connections after mobility, showing that scientists who change affiliations could retain their connections and social capital established in the former host institutions or host countries. Scientists often belong to external social and professional communities that provide access to resources based on collaboration (Almeida, Hohberger, & Parada, 2011). Given these, social capital scientists constructed in a previous affiliation could be transported to the new affiliation as they still belong to the same community after they move (Tartari et al., 2018). Long-distance collaboration activities increased with recent advances in information and telecommunication technologies (Frenken, Hardeman, & Hoekman, 2009). In this case, prior scientific connections could endure over time, space and organizational boundaries. Existing literature provides supporting evidence of this argument and suggests that moving to a new affiliation does not necessarily mean a loss of prior social and professional ties that have already been established (Jonkers & Cruz-Castro, 2013; Jonkers & Tijssen, 2008; Melkers & Kiopa, 2010; Scellato et al., 2015; Velema, 2012; Wang, Hooi, Li, & Chou, 2019).

Changing affiliations enables scientists to initiate new connections since social relationships are more likely to be formed through co-location. Considering this fact, some researchers believe that scientists' mobility leads to a net increase in social or professional capital, which could improve post-mobility productivity. This argument is supported by empirical evidence regarding domestic mobility (Dokko & Rosenkopf, 2010; Somaya, Williamson, & Lorinkova, 2008). Furukawa, Shirakawa, and Okuwada (2011) suggest that international mobility may facilitate collaboration, which is evidenced by overlaps in collaborative and mobility networks.

### 2.3. Scientific stratification and the effect of mobility

It is unclear whether and how upward mobility and non-upward mobility are associated with scientists' post-mobility productivity. Science is stratified since resources are highly concentrated in a few individuals and prestigious institutions (Davies & Zarifa, 2012; Merton, 1968; Salmi, 2010). Elite institutions could provide better access to ideas, equipment, high-quality social capital and other resources. Joining a higher-ranked institution might be accompanied by more scientific resources, higher institutional prestige and high-quality colleagues, and thus might improve scientists' productivity and collaboration network. Transitioning to a higher-ranked university not only means upward social mobility (Fernández-Zubieta et al., 2015, 2016), but also enables scientists to be embedded in a better research environment in which their productivity could be improved more (Martin-Rovet, 2003). The existing literature suggests inconsistent findings of the impact of mobility's direction on scientific productivity (Deville et al., 2014; Fernández-Zubieta et al., 2016)

### 2.4. Domestic mobility and international mobility

Most prior studies on mobility investigate scientists' mobility across national borders with the increasing globalization of science, which is referred to as international mobility. Accordingly, domestic mobility is referred to as the movement of scientists across institutions or regions within a country (Netz et al., 2020). The distinction between domestic and international mobility in various aspects makes the barriers scientists face and the benefits they obtain different.

Domestic mobility cannot generate transnational capital or international academic links that can be obtained through international mobility. A growing body of literature has empirically investigated the links between international mobility and scientists' performance measured by various indicators (Fontes, Videira, & Calapez, 2013; Jonkers & Tijssen, 2008; Scellato, Franzoni, & Stephan, 2012). Most previous literature attributes the positive effect of international mobility to the accumulated transnational professional ties and knowledge spillovers from their colleagues in the former host country. Insights of the knowledge recombination theory suggest that the combination of knowledge from distant resources is often associated with greater idea generation and innovation (Franzoni, Scellato, & Stephan, 2014; Hargadon & Sutton, 1997). International mobility facilitates scientists' collaboration with peers

in the cross-country context and enhances their learning of more diverse and distant knowledge (Jonkers & Tijssen, 2008). Those benefits that could be better achieved through international mobility, relative to domestic mobility, can improve scientists' career development, presumably to a larger extent.

International mobility is more constrained and generate transnational costs due to long-distance moving, restrictions of immigration policies (Gaughan & Robin, 2004), cultural shocks (Bhugra, 2003) and the differences in research system and academic labor market between the source country and the destination country (Shin & Jung, 2014). These factors might increase the costs of adaptations, the degree to which scientists' routine and social networks are disrupted, and thus hinder scientists' productivity and collaboration. International mobility makes scientists further away from their local professional network constructed in their previous institutions. Therefore, it is more challenging for scientists who experienced international mobility to maintain collaborative relationships with their former colleagues over a long distance (Audretsch & Stephan, 1996; Ponds, Van Oort, & Frenken, 2007). In this sense, the professional ties scientists established in the former institution should be more enduring for domestic mobility, relative to international mobility. The research system, academic job market and work environment vary across countries so do the candidate selection criteria, salaries, rewards and workload (Shin, Arimoto, Cummings, & Teichler, 2014; Shin & Jung, 2014). Those discrepancies make scientists spend more time adapting to the new political, cultural and academic environment after they moved to another country (Gill, 2010). Besides, prior literature notes that scientists might be discriminated against if they have overseas experiences and returned to their country of origin (Ackers, 2008). In short, those detriments caused by international mobility might slow down the pace of scientists' career development. A recent empirical study provides evidence of the negative effect of international mobility on the late-phase career advancement of researchers (Li & Tang, 2019).

Given the above distinctions between the two types of mobility, the findings regarding the effect of international mobility might not be consistent with those of domestic mobility. This study focuses on domestic mobility that is scant in prior literature and explores the effect of domestic mobility on scientists' collaboration and productivity.

### 2.5. Research questions

To fill the research gaps discussed earlier, focusing on domestic mobility, we propose the following research questions:

RQ1: Does mobility spur or hamper scientists' productivity?

RQ2: Does mobility influence acquiring new and preserving previous collaboration partners?

RQ3: Do upward mobility and non-upward mobility differ in how they influence scientists' productivity and collaboration?

## 3. Data and methodology

### 3.1. Sample construction and variables

In Scopus database, 517,775 publications with at least one Chinese address are assigned to the category of computer science (CS) before 2013. Specifically, we define a publication that is divided into any ASJC (All Science Journal Classification) code[1] between 1700 and 1712 as a CS publication. These publications are authored by 566,060 distinct authors of which 90.65% are affiliated with a Chinese address (denoted as CS authors hereafter). We extract all the publication records of CS authors from Scopus and keep those with at least 10 publications[2] before 2013,[3] with 105,227 authors remaining. This study only keeps the authors with a total productivity of no fewer than 10 papers. Those sampled scientists are nearly the top 20% productive authors among Chinese scientists who published at least one CS paper. This is because the mobility records for less prolific researchers may be less accurate since the change of affiliations may not be reflected in their publication records. It is noteworthy that these publications may not necessarily be in the CS category, as some of the CS authors may publish in other disciplines. The selection of productivity threshold is subjective as there is no standard criterion. To ensure that the threshold of 10 will not bias the results, we change the threshold to 5 for a robustness check. Generally, the major findings are consistent. Please see Tables A9 and A11 in Supplementary Information (SI). We keep authors with the address information containing either "univ" or "coll" or "instit" and match the affiliations of authors to the corresponding Chinese names of the universities,[4] and thus 63,882 authors remain.[5] In Scopus, each publication is assigned

[1] ASJC codes represent the scientific discipline of the journals in which the article was published, and are provided by Scopus. A publication can be assigned several ASJC codes. Please see the link: https://service.elsevier.com/app/answers/detail/a_id/15181/supporthub/scopus/

[2] It is challenging to fully capture the changes of affiliations for scientists with low productivity.

[3] The dataset we have is up to 2013. Due to the large sample size and the sufficient time windows, the study periods between 2000 and 2012 may not hamper the generalization of the findings.

[4] We match the latest version of Chinese names with the university that was merged into another university or changed names. For example, Zhejiang University is established as a combination of four universities in Hangzhou including Zhejiang University, Hangzhou University, Zhejiang Agricultural University and Zhejiang Medical University. These four former versions are considered as Zhejiang University.

[5] There are several reasons why we only consider scientists in universities. Scientists in universities produce a major proportion of publications. More than 81% of author-publication pairs with address information where either "univ" or "coll" or "institi" is included. One big challenge to capture the mobility data of scientists is to identify scientists' universities from universities' names indexed in Scopus which are usually not standard due to translation from Chinese names to English names. Besides, some institutions changed names, were merged or separated. To address these problems, we manually match universities' English names to their Chinese names by using publicly available information. However, for non-university institutions, this would be more challenging as their official English names may not be available online or the information about the history (such as name changing, separating or merging) of these institutions is not sufficient.

ASJC codes that indicate the discipline the publication belongs to.[6] We consider the discipline that accounts for the largest share of an authors' publications as the major discipline of the author. The authors in the disciplines that account for less than 1% of the total sampled authors are removed. The final dataset includes 62,330 Chinese authors. The distribution of the major disciplines the sampled scientists worked in, and the distributions of their total productivity and start year which is defined as the year when the first publication was published, are shown in Figures A1 to A2 in SI, respectively.

Scopus automatically distinguishes between authors sharing the same names and matches variations of author names by assigning a unique identifier (i.e., Scopus author ID) to each author. Several studies assessed the accuracy of Scopus author ID using external information (e.g., funding databases and CV data), and concluded that it is a good tool to track scientific mobility if no CV data is available (Aman, 2018; Kawashima & Tomizawa, 2015). [7]

This study only keeps authors' affiliations listed as the first when authors have multiple affiliations. We suppose it does not change the major findings in this study because scientists usually list their primary affiliations as the first affiliations in their papers due to their heavy dependence on the affiliations they serve in China. We find that only 5.4% of scientists were affiliated with multiple institutions in the dataset. In addition, it should be noted that papers with affiliations in foreign countries are excluded, which means that international mobility, i.e., mobility between a foreign affiliation and a Chinese affiliation, will not be considered in this study. Different from most prior literature that investigates international mobility, this study highlights its contribution to enriching the knowledge of domestic mobility that is often missing in existing literature. The distinctions between the two types of mobility that are discussed in section 2.4, make the barriers scientists face and the benefits they obtain different, and thus the types of mobility might influence scientists' collaboration and productivity differently.

To address the first research question (RQ1), our primary research objective is to measure the effect of domestic mobility on scientists' productivity in emerging economies. We use the number of publications of scientists in a year (*publication*), and the average number of citations in a five-year citation window (i.e., within five years since the paper is published) per publication received by those publications (*citation*), as the main dependent variables, which are usually used to measure scientific productivity (Azoulay et al., 2017; Fernandez-Zubieta et al., 2013; Liu, Shi, & Li, 2017; Xu et al., 2022; Bu, Li, Wei, Liu, & Li, 2020). To capture whether and how mobility influences scientists' collaboration (RQ2), we introduce two sets of dependent variables: *new collaborator* to represent the number of distinct collaborators in the year with whom the scientists never collaborated before, and *previous collaborator* to indicate the number of distinct collaborators in the year with whom the scientists had collaborated in the past; *new collaborator (ratio)* which is defined as the number of new collaborators divided by the total distinct co-authors in the year, and *previous collaborator (ratio)* that is defined as the number of previous collaborators divided by the total distinct co-authors in the year. The first set of dependent variables concerning collaboration is used to measure the absolute number of scientists' new and previous collaborators; the second set of dependent variables is used to measure scientists' tendency to construct new collaboration or maintain repeat collaboration.

The major explanatory variable is whether the scientist has changed an affiliation in the year (*mobility*). Consistent with previous literature (Robinson-Garcia et al., 2019), we find that changing affiliations rarely takes place for Chinese scientists, with 15.9% of scientists moving to a new affiliation and generating 10,429 mobility events from 2000 to 2012. 71.01% of movers experienced a job change only once or twice.

We generate two additional explanatory variables by differentiating upward/downward direction of mobility to address RQ3 due to the importance of the research environment and reputation factors in shaping scientists' productivity. All Chinese universities could be classified into four categories based on universities prestige (Allen, 2017; Zong & Zhang, 2019), C9 League universities,[8] 985 universities (excluding C9 universities),[9] 211 universities (excluding 985 universities)[10] and other ordinary universities. C9 League universities are often referred to as China's Ivy League, which are considered the top-tier universities in China. 985 and 211 universities are second-tier and third-tier universities, respectively. The remaining ordinary universities are the fourth-tier universities in China. Upward mobility (*upmobility*) is defined as scientists' move to a higher-tier university that accounts for 32.07% of the total mobility events, and non-upward mobility (*nupmobility*) refers to mobility to a lower-tier university or a university of the same tier. It should be noted that the scientists at C9 universities are removed when the upward mobility is examined since it was impossible for them to move any higher.

A set of control variables are included to control for factors that may be related to scientists' research performance. We include (1) scientists' academic age (*age*) defined as the number of years since scientists' first publications to account for potential life cycle effects (Jones & Weinberg, 2011; Levin & Stephan, 1991); (2) the reputation of scientists' affiliation to control for research environment, organizational reputation and peer effects, measured by whether the scientist is affiliated with an elite university (*elite*), that is defined as a university either in the first, second or third tiers, in the last year (Fernandez-Zubieta et al., 2013; Long, Allison, & McGinnis, 1993); (3) the number of publications published in the past three years (*publication_past3*), and the average citations per publication (*citation_past3*) of these publications in a five-year citation window to control for scientists' past productivity (Azoulay et al., 2017; Fernandez-Zubieta et al., 2013); (4) the number of distinct collaborators in the past three years (*collaborator_past3*) to measure

[6] One publication can have multiple ASJC codes.

[7] To further verify the accuracy of Scopus author IDs of Chinese authors, we manually check 15 Chinese scholars in the CS field by comparing their publications listed in Scopus and on their homepages or Google Scholar profiles linked from their homepages. The total average precision and recall are 0.95 and 0.85 respectively. As publications on homepages may not be complete, and some conference papers are not indexed in Scopus, the true precision and recall should be higher and thus are acceptable for this study. Details of this check are presented in Table A12 in SI.

[8] http://www.moe.gov.cn/jyb_xwfb/s5147/201611/t20161101_287192.html

[9] http://www.moe.gov.cn/srcsite/A22/s7065/200612/t20061206_128833.html

[10] http://www.moe.gov.cn/srcsite/A22/s7065/200512/t20051223_82762.html

**Table 1**
Summary statistics of the sampled scientists (N=62,330).

| Variable | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|
| **Dependent variable** | | | | |
| Publication | 1.178 | 0.787 | 0 | 4.477 |
| Citation | 0.749 | 0.88 | 0 | 6.956 |
| Previous collaborator | 1.051 | 0.864 | 0 | 8.305 |
| New collaborator | 1.165 | 1.029 | 0 | 8.383 |
| Previous collaborator (ratio) | 0.477 | 0.331 | 0 | 1 |
| New collaborator (ratio) | 0.523 | 0.331 | 0 | 1 |
| **Independent variable** | | | | |
| Mobility | 0.063 | 0.244 | 0 | 1 |
| Upmobility | 0.02 | 0.139 | 0 | 1 |
| Nupmobility | 0.044 | 0.204 | 0 | 1 |
| **Control variable** | | | | |
| Age | 1.666 | 0.793 | 0 | 3.135 |
| Elite | 0.86 | 0.347 | 0 | 1 |
| Publication_past3 | 1.938 | 0.927 | 0 | 4.736 |
| Citation_past3 | 1.449 | 1.186 | 0 | 7.426 |
| Collaborator_past3 | 2.531 | 1.13 | 0 | 9.291 |
| ICpaper | 0.104 | 0.32 | 0 | 4.382 |
| Teamsize | 0.872 | 0.46 | 0 | 4.635 |

**Notes:** variables except elite, previous collaborator (ratio), new collaborator (ratio) and mobility variables, are naturally log-transformed.

**Table 2**
Summary statistics of movers and stayers.

| | Mover(N=63,592) | | Stayer(N=476,522) | |
|---|---|---|---|---|
| Variable | Mean | Std. Dev. | Mean | Std. Dev. |
| Publication | 1.219 | 0.808 | 1.173 | 0.784 |
| Citation | 0.794 | 0.904 | 0.743 | 0.877 |
| Previous collaborator | 1.05 | 0.906 | 1.051 | 0.858 |
| New collaborator | 1.315 | 1.116 | 1.145 | 1.016 |
| Previous collaborator (ratio) | 0.43 | 0.332 | 0.483 | 0.331 |
| New collaborator (ratio) | 0.57 | 0.332 | 0.517 | 0.331 |
| Age | 1.701 | 0.781 | 1.662 | 0.795 |
| Elite | 0.821 | 0.383 | 0.865 | 0.342 |
| Publication_past3 | 1.974 | 0.939 | 1.933 | 0.925 |
| Citation_past3 | 1.539 | 1.209 | 1.437 | 1.182 |
| Collaborator_past3 | 2.645 | 1.18 | 2.516 | 1.122 |
| ICpaper | 0.099 | 0.302 | 0.105 | 0.322 |
| Teamsize | 0.847 | 0.441 | 0.875 | 0.463 |

**Notes:** variables except elite, previous collaborator (ratio), new collaborator (ratio) and mobility variables, are naturally log-transformed.

scientists' recent social capital that may influence their productivity (Bozeman & Boardman, 2014); (5) the number of international collaborated papers (*ICpaper)* published in the year to measure the degree to which scientists participate in international collaboration (Liu et al., 2021; Wagner, Whetsell, & Mukherjee, 2019); (6) and the average number of authors (*teamsize*) for papers scientists publish in the year to measure the average team size of scientists' papers (Lariviere, Sugimoto, Tsou, & Gingras, 2014).

The definition of variables is shown in Table A1 in SI. The summary statistics and correlation matrix of variables are presented in Table 1 and Table A2 in SI, respectively. Table 2 reports summary statistics for movers and unmoved scientists, suggesting that movers are more productive and elder than their static peers.

### 3.2. Methodology

In this section, using a PSM-DID estimator, we first explore the effect of mobility on productivity and collaboration to address RQ1 and RQ2. Differentiating the nature of mobility (i.e., upward mobility and non-upward mobility), we investigate RQ3.

#### 3.2.1. Baseline models: Difference in differences strategy

A baseline model is constructed based on a difference in differences (DID) strategy (Liu et al., 2022). Following existing literature (Fernandez-Zubieta et al., 2013), we assume that mobility has a lasting career impact on scientists' productivity and collaboration. The independent variable is $mobility_{it}$, which refers to whether or not scientist *i* has changed from a domestic affiliation to another domestic affiliation in year *t*. The independent variable equals one for all the years following the first move of scientist *i* and zero otherwise.

**Table 3**
The PSM-DID estimates of the effects of scientists' mobility on productivity and collaboration around the move year (N=79,984, pairs=7,071).

| | (1) Publication | (2) | (3) Citation | (4) | (5) Previous collaborator | (6) | (7) New collaborator | (8) | (9) Previous collaborator(ratio) | (10) | (11) New collaborator(ratio) | (12) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mobility | **0.098**$^{***}$ | | **0.021** | | **0.032**$^{**}$ | | **0.174**$^{***}$ | | **-0.033**$^{***}$ | | **0.033**$^{***}$ | |
| | (0.012) | | (0.013) | | (0.013) | | (0.023) | | (0.007) | | (0.007) | |
| t-3 | | -0.027 | | -0.033 | | -0.023 | | -0.053$^{*}$ | | 0.008 | | -0.008 |
| | | (0.016) | | (0.025) | | (0.019) | | (0.026) | | (0.006) | | (0.006) |
| t-2 | | 0.019 | | 0.006 | | 0.005 | | 0.010 | | 0.006 | | -0.006 |
| | | (0.018) | | (0.019) | | (0.019) | | (0.026) | | (0.008) | | (0.008) |
| t-1 | | 0.026$^{*}$ | | 0.025 | | 0.002 | | 0.037 | | -0.001 | | 0.001 |
| | | (0.013) | | (0.015) | | (0.012) | | (0.022) | | (0.007) | | (0.007) |
| t0 | | 0.130$^{***}$ | | 0.052$^{***}$ | | 0.127$^{***}$ | | 0.137$^{***}$ | | 0.013$^{**}$ | | -0.013$^{**}$ |
| | | (0.011) | | (0.016) | | (0.012) | | (0.018) | | (0.006) | | (0.006) |
| t+1 | | 0.070$^{***}$ | | 0.023$^{*}$ | | -0.015 | | 0.121$^{***}$ | | -0.030$^{***}$ | | 0.030$^{***}$ |
| | | (0.011) | | (0.012) | | (0.015) | | (0.017) | | (0.006) | | (0.006) |
| t+2 | | 0.054$^{***}$ | | 0.008 | | -0.005 | | 0.078$^{***}$ | | -0.019$^{***}$ | | 0.019$^{***}$ |
| | | (0.010) | | (0.014) | | (0.015) | | (0.014) | | (0.005) | | (0.005) |
| t+3 | | 0.030$^{***}$ | | -0.010 | | -0.024$^{*}$ | | 0.076$^{***}$ | | -0.025$^{***}$ | | 0.025$^{***}$ |
| | | (0.009) | | (0.013) | | (0.013) | | (0.016) | | (0.006) | | (0.006) |
| Controls | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included |
| Individual fixed effects | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included |
| Year fixed effects | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included |
| Constant | 0.666$^{***}$ | 0.964$^{***}$ | 0.088 | 0.232$^{**}$ | -0.412$^{***}$ | -0.345$^{***}$ | 0.611$^{***}$ | 0.733$^{***}$ | 0.200$^{***}$ | 0.183$^{***}$ | 0.800$^{***}$ | 0.817$^{***}$ |
| | (0.048) | (0.049) | (0.087) | (0.077) | (0.048) | (0.042) | (0.072) | (0.066) | (0.017) | (0.017) | (0.017) | (0.017) |
| R-squared | 0.554 | 0.554 | 0.517 | 0.525 | 0.646 | 0.649 | 0.503 | 0.507 | 0.317 | 0.318 | 0.317 | 0.318 |

**Notes:** in columns 1, 3, 5, 7, 9 and 11, the independent variable (*Mobility*) is whether scientists have changed an affiliation in the year; in columns 2, 4, 6, 8,10, and 12, t0 refers to the year of move, and t-n/t+n indicate n years prior to/after the year of mobility; controls include variables shown in Table A1 in SI and are not displayed; the dependent variables are the number of papers scientists published in the year (columns 1 to 2), the average number of citations in a five-year citation window per publication received by scientists' publications published in the year (columns 3 to 4), the number of distinct co-authors scientists have collaborated with in the past (columns 5 to 6), the number of distinct co-authors scientists have never collaborated with in the past (columns 7 to 8), the fraction of distinct previous collaborators to the total number of distinct co-authors in the year (column 9 to 10), and the fraction of distinct new collaborators to the total number of distinct co-authors in the year (column 11 to 12); fixed effects regarding year are incorporated to control time-variant unobserved changes, and individual fixed effects are included to control scientists' time-invariant characteristics; robust standard errors are in parentheses; $^{***}$, $^{**}$, and $^{*}$ represent significance at the 1%, 5%, and 10% level.

The coefficients for $mobility_{it}$ are before-after estimates of the impact of mobility on scientists' productivity and collaboration. A set of control variables that are shown in Table 1 are considered. Time-fixed effects, $Y_t$, are incorporated to control time-variant unobserved changes, such as the policy changes in China's scientific system. Individual fixed effects, $S_i$, are used to control scientists' time-invariant unobservable characteristics, such as ability, attitude and other personal traits. The relationship between scientists' mobility and their productivity and collaboration is estimated by Eq. 1. An OLS linear model is applied to estimate Eq. 1.

$$Dependent\ variable_{it} = \alpha + \beta Mobility_{it} + Controls + Y_t + S_i + \varepsilon \tag{1}$$

We further differentiate upward move (i.e., moving to a higher-tier university) and non-upward move (i.e., moving to a lower-tier university or a university of the same tier) to see whether the direction of mobility matters for the effect of mobility. In Eq. 1, we replace $mobility_{it}$ with $upmobility_{it}$ that equals one if scientists have moved to a higher-tier university, and with $nupmobility_{it}$ that equals one if scientists have moved to a lower-tier university or a same-tier university.

To dig deeper into the dynamic effect of mobility, following Agrawal (2013), we replace the mobility variable in Eq. 1 with seven dummies variables that relate scientists' productivity and collaboration to the mobility event around the mobility year:

$$Dependent\ variable_{it} = \alpha + \beta_1(t-3)_{it} + \beta_2(t-2)_{it} + \beta_3(t-1)_{it} + \beta_4 t0_{it} + \beta_5(t+1)_{it} + \beta_6(t+2)_{it} + \beta_7(t+3)_{it} + Controls + Y_t + S_i + \varepsilon \tag{2}$$

Where $t0_{it}$ relates the year when scientist $i$ moved to a new affiliation; $(t-n)_{it}$ indicates $n$ year(s) before scientist $i$ changed to a new affiliation ; $(t+n)_{it}$ refers to $n$ year(s) after scientist $i$ changed to a new affiliation. A mobility event that causes an effect in a given year can only occur in the year of the mobility event or later years, i.e., *t0* and *t+n*. Similarly, we introduce seven dummy variables for $upmobility_{it}$ and $nupmobility_{it}$ to capture the dynamic effect of two types of mobility events on scientists' productivity and collaboration after the move.

Although DID is effective in estimating the outcome that is attributable to the intervention (scientists' mobility in this paper), it has an assumption that changes in time-variant factors in treatment and control groups are equal (known as the Parallel/Common Trend Assumption). Otherwise, the effect of intervention could be overestimated or underestimated. In this paper, this assumption is satisfied only if the dependent variables of the two groups have parallel trends in the absence of scientists' mobility.

### 3.2.2. Counterfactual framework: PSM-DID estimator

*Propensity score matching.* The investigation of whether mobility impacts scientists' productivity and collaboration should take selection bias into account. Scientists' mobility and collaboration are driven by multiple factors, such as scientists' academic ability and other characteristics. Previous literature has documented that productive scientists are more likely to move (Azoulay et al., 2017; Ganguli, 2015; Lenzi, 2009; Zucker, Darby, & Torero, 2002). Accordingly, the higher productivity of moved scientists after they move may result from their stronger academic ability rather than their job transition, and thus it is not sufficient to make a simple comparison of productivity between movers and unmoved scientists. A more interesting question is whether movers would perform better if they did not move to a new affiliation.

Propensity score matching (PSM) is a widely used technique for estimating the average treatment effect by mitigating selection bias based on observable variables. It is an approach where treated and untreated units are matched on the estimated probability of receiving treatment based on their observable characteristics, and thus creating a comparable counterfactual (Liu et al., 2018; Liu & Hu, 2021; Rosenbaum & Rubin, 1983). The advantage of PSM is to adjust for imbalance in observed variables and control for the possible ex-ante selection bias.

Though PSM could mitigate selection bias concerning scientists' mobility based on observable variables (i.e., career age, past productivity and collaboration, discipline and the tier of affiliations scientists belong to in this study), it cannot address selection bias caused by unobservable variables that are not included in this study, such as gender, ambition, attitude, career planning and research potential. Therefore, this method cannot guarantee that the unobservable variables are balanced between treated groups and untreated groups.

*PSM-DID estimator.* The combination of PSM and DID (Heckman, Ichimura, & Todd, 1997), i.e., the PSM-DID estimator, could overcome the problem of selection on both observables and unobservables to capture unbiased causal effects. Matching estimators are often ineffective due to the strong assumption of selection on observables. DID estimator is sometimes unsatisfactory because of the violation of the assumption of common trend (Dehejia, 2005). The combination of PSM and DID could offset their individual weaknesses. On the one hand, in contrast to the traditional DID, the PSM-DID estimator reduces selection bias by comparing changes in moved scientists' productivity and collaboration and those for observationally similar control scientists, which helps satisfy the common trend assumption. The matching procedure ensures that the only remaining difference between the treatment and control groups is whether or not a scientist moved (Blundell & Dias, 2009). On the other hand, the PSM-DID estimator controls for unobserved time-invariant differences across moved scientists and unmoved ones, which mitigates the problem of selection bias caused by unobservable variables if only PSM is used (Cadot, Fernandes, Gourdon, & Mattoo, 2015; Heckman et al., 1997). In one of the first empirical applications of the PSM-DID technique, researchers used PSM, DID and PSM-DID estimators to estimate the effect of a labour market training program and found that the PSM-DID technique was the most robust and effective in estimating the causal effect (Smith & Todd, 2005). Given the above virtues, the PSM-DID estimator has been increasingly used to improve the quality of non-experimental evaluation results in recent decades (Blundell & Costa Dias, 2000; Girma & Görg, 2007; Smith & Todd, 2005).

We use the PSM-DID estimator to address selection on both observables and unobservables. There are two steps to perform the PSM-DID analysis. First, we estimate a nearest neighbor propensity score model by performing 1-to-1 nearest-neighbor PSM

without replacement using psmatch2 package in Stata. This procedure compares movers, or scientists who experienced upward move (*upward movers*), or scientists who changed affiliations to a university of the same tier or a lower-tier university (*non-upward movers*), to their nearest neighbors who never moved throughout the study period and have the similar personal and academic characteristics with movers in the year when movers experienced a mobility event. The propensity score of observations is estimated by constructing a logistic regression model where treatment status is estimated based on the baseline characteristics including career age, past productivity and past collaboration in the year. Among a pool of unmoved scientists with the closest propensity score, a moved scientist is matched to an unmoved one with the same discipline and the same tier of their affiliation in the last year. We end up with the matched sample including 7,071 matched pairs of moved scientists and their corresponding unmoved peers. 2,131 upward moved scientists, and 4,486 non-upward moved scientists are matched to their nearest neighbors, respectively. Covariates balance checking is shown in SI (Table A3 to Table A8 and Figure A3). Subsequently, we employ a DID estimator that is defined in Eq. 1 to capture the effect of scientists' mobility on their productivity and collaboration based on the panel data on movers, up-ward movers, non-upward movers and their matched unmoved pairs, separately. The inclusion of individual fixed effects could help remove possible selection bias caused by time-invariant unobservable differences across groups.

## 4. Results

### 4.1. The relationship between scientists' mobility and productivity

The results of the baseline models concerning the effect of mobility on scientists' productivity and collaboration are shown in Fig. 1. Fig. 1 suggests that in some years, the Parallel Trend Assumption is violated. Before the intervention (t0), the differences in the number of publications between treatment and control groups generally grow over time, which is evidenced by the increasingly larger coefficients over years before t0. For example, in t-3 and t-1, the coefficients of mobility on publications are significant. This implies that moved scientists and unmoved ones do not have parallel trends regarding their publications before the mobility event, and thus DID models might lead to inaccurate estimates.

We perform DID analyses with the PSM method to acquire more accurate estimates of the effect of mobility on scientists' productivity. Based on the matched samples using PSM, generally, there are no significant differences between moved scientists and their matched unmoved peers in the number of publications and citations regardless of the direction of mobility in the three years preceding the mobility event (in columns 2 and 4 of Tables 3 to 5, coefficients of t-3 and t-2 are insignificant and close to zero). Fig. 2 shows no indication of a pre-existing trend that affects scientists' productivity before the year of move (e.g., t-3, t-2 and t-1). This is because in most of the years before t0, moved scientists and their matched unmoved peers do not significantly differ from each other in publication and citation. [11] The above evidence reflects that the common trend assumption is generally met.

Based on the panel data on the matched sample, we find that after the move, the number of publications for mobile scientists is significantly higher than that for unmoved matched peers by 9.8% (coefficient: 0.098, $p<0.01$) (see column 1 of Table 3) since the move year. Specifically, the number of publications for treatment scientists is significantly higher than that for unmoved matched pairs by 13%, 7%, 5.4% and 3% in the year of move, one year, two years and three years after the move year, respectively (see column 2 of Table 3). Generally, we observe insignificant differences in citations between treatment scientists and matched untreated scientists, which is evidenced by the insignificant coefficient of *mobility* on citation (column 3 of Table 3).

Differentiating the upward/downward direction of mobility, we observe a significantly positive impact of scientists' upward mobility and that of non-upward mobility on publication counts. After the move, upward movers have 10% (coefficient: 0.100, $p<0.01$) more publications than their matched unmoved peers (see column 1 of Table 4). Non-upward movers published 9.2% (coefficient: 0.092, $p<0.01$) more publications than their matched unmoved colleagues (see column 1 of Table 5). Again, we observe insignificant effects of upward (see column 3 of Table 4) and non-upward mobility (see column 3 of Table 5) on citations.

The dynamic estimated effects of scientists' mobility on productivity in and after the move year are illustrated in Fig. 2 and reported in columns 2 and 4 of Tables 3 to 5. The plots indicate a clear increased number of publications for moved scientists in and after the mobility event regardless of the direction of the move, which is evidenced by the significantly positive coefficients of t0 and t+n. However, the increase in citations is slight and is insignificant in most of the years after the mobility. The finding might suggest the irrelevance between mobility and scientists' citations. It is worthy of note that the observed increase in publications in the move year is the sharpest, relative to the increase in the subsequent years, which indicates that the effect of mobility is immediate and decaying with time.

### 4.2. The relationship between scientists' mobility and collaboration

Again, the baseline models might generate unreliable estimates of the effect of mobility on scientists' collaboration. As shown in Fig. 1(D) to (I), in some years before t0, moved scientists and unmoved ones are significantly different in the dependent variables regarding new collaborators or previous collaborators. For example, in Fig. 1(D), before t0, the difference between moved scientists and unmoved scientists in their new collaborators shows an upward trend due to the increasingly larger coefficients from t-3 ($p<0.01$) to t-2 ($p<0.1$).

[11] The coefficients of t-1 on publication in Tables 3 and 4 are significant at the significance level of 10%, while their p-values are 0.06 and thus the coefficients of t-1 are nearly insignificant.

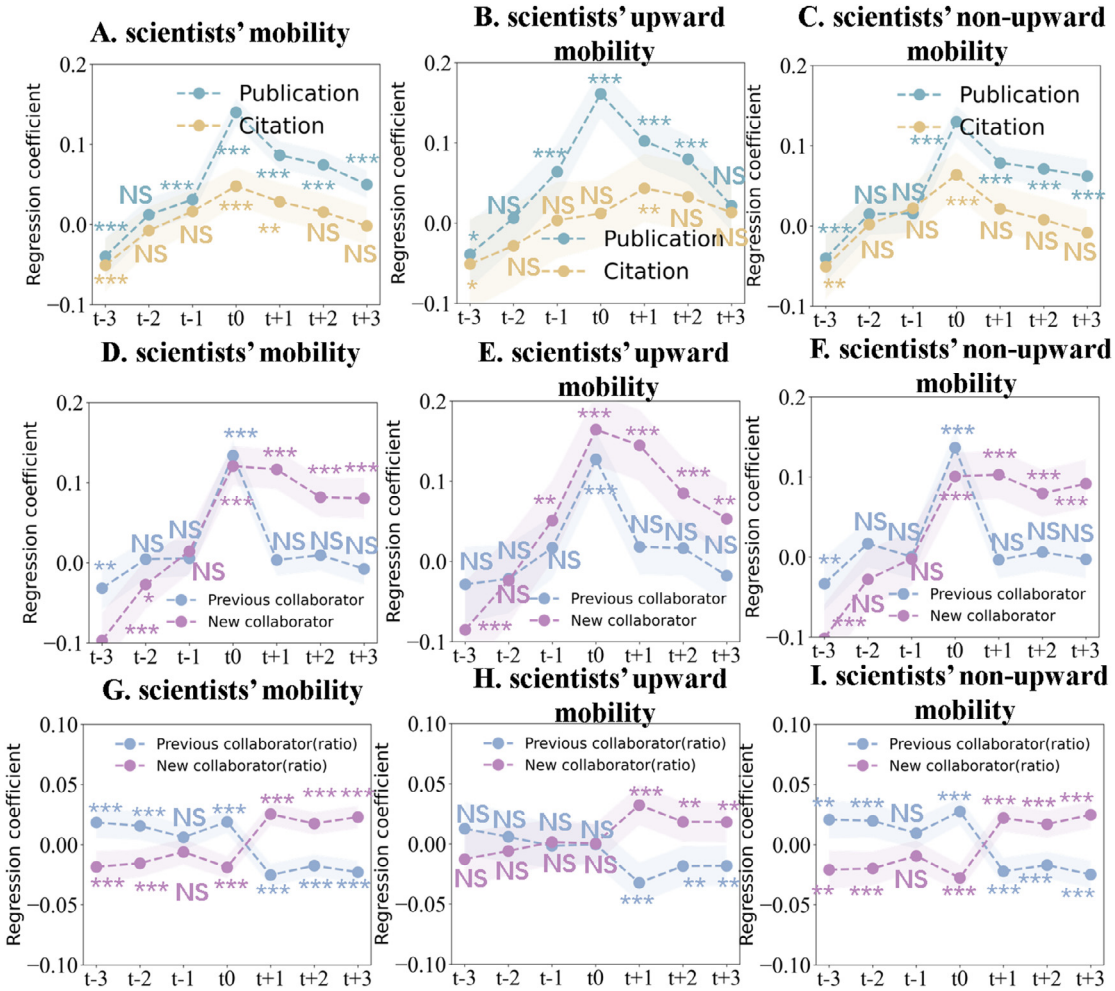


**Fig. 1.** The DID estimates of mobility on scientists' productivity and collaboration around the move year without PSM.
**Notes:** The shaded area represents ± 1.96 * std. error of each point estimate; t-n/t+n indicates n years prior to/after the year of mobility, and t0 refers to the year of move; the coefficient means the difference between treated scientists and their matched untreated scientists in the dependent variables; ***, **, and * represent significance at the 1, 5, and 10% levels and NS stands for not significant.

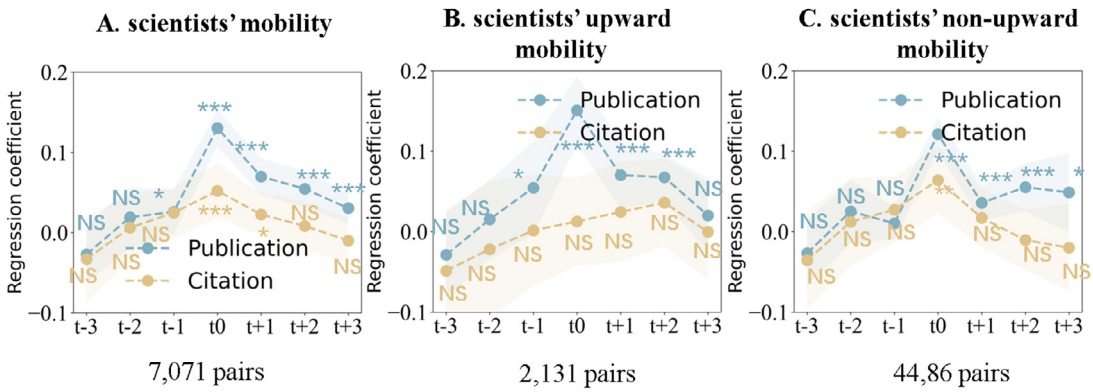


**Fig. 2.** The PSM-DID estimates of the effects of scientists' mobility on productivity around the move year.
**Notes:** The shaded area represents ± 1.96 * std. error of each point estimate; t-n/t+n indicate n years prior to/after the year of mobility, and t0 refers to the year of move; the coefficient means the difference between treated scientists and their matched untreated scientists in the dependent variables; ***, **, and * represent significance at the 1, 5, and 10% levels and NS stands for not significant.

**Table 4**
The PSM-DID estimates of the effects of scientists' upward mobility on productivity and collaboration around the move year (N=26,662, pairs=2,131).

| | (1) Publication | (2) | (3) Citation | (4) | (5) Previous collaborator | (6) | (7) New collaborator | (8) | (9) Previous collaborator(ratio) | (10) | (11) New collaborator(ratio) | (12) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Upmobility | **0.100**$^{***}$ | | **0.013** | | **0.053**$^{***}$ | | **0.170**$^{***}$ | | **-0.029**$^{***}$ | | **0.029**$^{***}$ | |
| | (0.019) | | (0.023) | | (0.016) | | (0.029) | | (0.008) | | (0.008) | |
| t-3 | | -0.029 | | -0.049 | | -0.008 | | -0.060 | | 0.009 | | -0.009 |
| | | (0.020) | | (0.035) | | (0.029) | | (0.042) | | (0.014) | | (0.014) |
| t-2 | | 0.014 | | -0.022 | | -0.003 | | -0.006 | | 0.003 | | -0.003 |
| | | (0.020) | | (0.039) | | (0.022) | | (0.042) | | (0.010) | | (0.010) |
| t-1 | | 0.054$^{*}$ | | 0.002 | | 0.035 | | 0.053 | | 0.000 | | -0.000 |
| | | (0.026) | | (0.032) | | (0.022) | | (0.041) | | (0.010) | | (0.010) |
| t0 | | 0.149$^{***}$ | | 0.013 | | 0.137$^{***}$ | | 0.163$^{***}$ | | -0.000 | | 0.000 |
| | | (0.019) | | (0.025) | | (0.020) | | (0.037) | | (0.008) | | (0.008) |
| t+1 | | 0.069$^{***}$ | | 0.024 | | 0.020 | | 0.129$^{***}$ | | -0.028$^{***}$ | | 0.028$^{***}$ |
| | | (0.021) | | (0.028) | | (0.023) | | (0.036) | | (0.009) | | (0.009) |
| t+2 | | 0.066$^{***}$ | | 0.036 | | 0.022 | | 0.082$^{**}$ | | -0.017$^{*}$ | | 0.017$^{*}$ |
| | | (0.013) | | (0.025) | | (0.025) | | (0.033) | | (0.009) | | (0.009) |
| t+3 | | 0.018 | | -0.000 | | -0.010 | | 0.039 | | -0.013 | | 0.013 |
| | | (0.023) | | (0.027) | | (0.022) | | (0.027) | | (0.008) | | (0.008) |
| Controls | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included |
| Individual fixed effects | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included |
| Year fixed effects | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included |
| Constant | 0.718$^{***}$ | 1.107$^{***}$ | 0.145 | 0.288$^{**}$ | -0.326$^{***}$ | -0.318$^{***}$ | 0.872$^{***}$ | 1.423$^{***}$ | 0.133$^{***}$ | 0.123$^{**}$ | 0.867$^{***}$ | 0.877$^{***}$ |
| | (0.087) | (0.021) | (0.128) | (0.113) | (0.097) | (0.100) | (0.134) | (0.064) | (0.043) | (0.044) | (0.043) | (0.044) |
| R-squared | 0.566 | 0.564 | 0.538 | 0.541 | 0.657 | 0.659 | 0.523 | 0.525 | 0.330 | 0.331 | 0.330 | 0.331 |

**Notes:** in columns 1, 3, 5, 7, 9 and 11, the independent variable (*Upmobility*) is whether scientists have changed to a higher-tier affiliation in the year; in columns 2, 4, 6, 8,10, and 12, t0 refers to the year of move, and t-n/t+n indicate n years prior to/after the year of mobility; controls include variables shown in Table A1 in SI and are not displayed; the dependent variables are the number of papers scientists published in the year (columns 1 to 2), the average number of citations in a five-year citation window received by scientists' publications published in the year (columns 3 to 4), the number of distinct co-authors scientists have never collaborated with in the past (columns 5 to 6), the number of distinct co-authors scientists have collaborated with in the past (columns 7 to 8), the fraction of distinct previous collaborators to the total number of distinct co-authors in the year (column 9 to 10), and the fraction of distinct new collaborators to the total number of distinct co-authors in the year (column 11 to 12); fixed effects regarding year are incorporated to control time-variant unobserved changes, and individual fixed effects are included to control scientists' time-invariant characteristics; robust standard errors are in parentheses; $^{***}$, $^{**}$, and $^{*}$ represent significance at the 1%, 5%, and 10% level.

**Table 5**
The PSM-DID estimates of the effects of scientists' non-upward mobility on productivity and collaboration around the move year (N=55,402, pairs=4,486).

| | (1) Publication | (2) | (3) Citation | (4) | (5) Previous collaborator | (6) | (7) New collaborator | (8) | (9) Previous collaborator(ratio) | (10) | (11) New collaborator(ratio) | (12) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Nupmobility | **0.092**$^{***}$ | | **0.015** | | **0.018** | | **0.147**$^{***}$ | | **-0.027**$^{***}$ | | **0.027**$^{***}$ | |
| | (0.013) | | (0.013) | | (0.015) | | (0.021) | | (0.007) | | (0.007) | |
| t-3 | | -0.027 | | -0.036 | | -0.029 | | -0.013 | | 0.012 | | -0.012 |
| | | (0.018) | | (0.028) | | (0.024) | | (0.013) | | (0.008) | | (0.008) |
| t-2 | | 0.025 | | 0.012 | | 0.012 | | -0.018 | | 0.010 | | -0.010 |
| | | (0.018) | | (0.025) | | (0.024) | | (0.012) | | (0.011) | | (0.011) |
| t-1 | | 0.010 | | 0.027 | | -0.009 | | 0.001 | | 0.003 | | -0.003 |
| | | (0.009) | | (0.019) | | (0.017) | | (0.008) | | (0.008) | | (0.008) |
| t0 | | 0.121$^{***}$ | | 0.064$^{***}$ | | 0.125$^{***}$ | | -0.041$^{***}$ | | 0.023$^{***}$ | | -0.023$^{***}$ |
| | | (0.010) | | (0.019) | | (0.015) | | (0.009) | | (0.007) | | (0.007) |
| t+1 | | 0.036$^{***}$ | | 0.017 | | -0.029 | | 0.037$^{***}$ | | -0.025$^{***}$ | | 0.025$^{***}$ |
| | | (0.010) | | (0.017) | | (0.018) | | (0.008) | | (0.007) | | (0.007) |
| t+2 | | 0.055$^{***}$ | | -0.010 | | -0.017 | | 0.024$^{**}$ | | -0.013$^{*}$ | | 0.013$^{*}$ |
| | | (0.014) | | (0.017) | | (0.018) | | (0.010) | | (0.007) | | (0.007) |
| t+3 | | 0.049$^{*}$ | | -0.020 | | -0.033$^{*}$ | | 0.037$^{**}$ | | -0.024$^{***}$ | | 0.024$^{***}$ |
| | | (0.023) | | (0.025) | | (0.017) | | (0.012) | | (0.006) | | (0.006) |
| Controls | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included |
| Individual fixed effects | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included |
| Year fixed effects | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included | Included |
| Constant | 0.632$^{***}$ | 1.079$^{***}$ | 0.210$^{**}$ | 0.295$^{***}$ | -0.352$^{***}$ | -0.296$^{***}$ | 0.661$^{***}$ | -0.179$^{***}$ | 0.211$^{***}$ | 0.195$^{***}$ | 0.789$^{***}$ | 0.805$^{***}$ |
| | (0.081) | (0.049) | (0.090) | (0.088) | (0.080) | (0.074) | (0.113) | (0.029) | (0.024) | (0.023) | (0.024) | (0.023) |
| Observations | 55,482 | 55,482 | 55,482 | 55,482 | 55,482 | 55,482 | 55,482 | 55,482 | 55,202 | 55,202 | 55,202 | 55,202 |
| R-squared | 0.561 | 0.575 | 0.529 | 0.535 | 0.653 | 0.656 | 0.513 | 0.886 | 0.324 | 0.325 | 0.324 | 0.325 |

**Notes:** in columns 1, 3, 5, 7, 9 and 11, the independent variable (*Nupmobility*) is whether scientists have changed to a lower-tier or same-tier affiliation in the year; in columns 2, 4, 6, 8,10, and 12, t0 refers to the year of move, and t-n/t+n indicate n years prior to/after the year of mobility; controls include variables shown in Table A1 in SI and are not displayed; the dependent variables are the number of papers scientists published in the year (columns 1 to 2), the average number of citations in a five-year citation window received by scientists' publications published in the year (columns 3 to 4), the number of distinct co-authors scientists have never collaborated with in the past (columns 5 to 6), the number of distinct co-authors scientists have collaborated with in the past (columns 7 to 8), the fraction of distinct previous collaborators to the total number of distinct co-authors in the year (column 9 to 10), and the fraction of distinct new collaborators to the total number of distinct co-authors in the year (column 11 to 12); fixed effects regarding year are incorporated to control time-variant unobserved changes, and individual fixed effects are included to control scientists' time-invariant characteristics; robust standard errors are in parentheses; $^{***}$, $^{**}$, and $^{*}$ represent significance at the 1%, 5%, and 10% level.

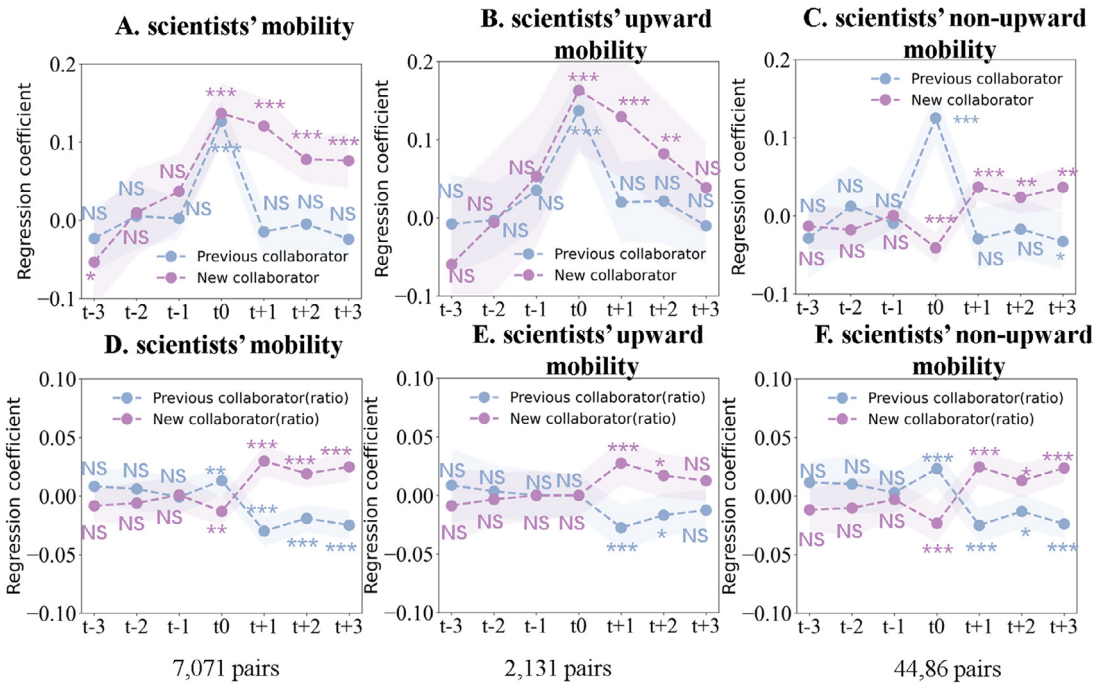


**Fig. 3.** The PSM-DID estimates of the effects of scientists' mobility on collaboration around the mobility year.
**Notes:** The shaded area represents ± 1.96 * std. error of each point estimate; t-n/t+n indicate n years prior to/after the year of mobility, and t0 refers to the year of mobility; the coefficient means the difference between treated scientists and their matched untreated scientists in the dependent variables; ***, **, and * represent significance at the 1, 5, and 10% levels and NS stands for not significant.

The parallel trend assumption is well satisfied based on the matched sample. Fig. 3(A)-(F) shows that regardless of the direction of mobility, moved scientists and their matched unmoved peers do not significantly differ from each other in the number of and the fraction of previous collaborators and that of new collaborators in the three years preceding the mobility year. This is evidenced by the insignificant coefficient of t-3, t-2 and t-1. Based on the panel data on the matched sample acquired using PSM, we conduct DID analyses and find that mobile researchers have 17.4% more new collaborators after the move compared to non-mobile matched pairs (coefficient: 0.174, p<0.01 in column 7 of Table 3). The number of previous collaborators mobile researchers have after the move are slightly higher than that of unmoved matched pair by 3.2% (coefficient: 0.032, p-value<0.05 in column 5 of Table 3). However, this effect is only observed in the year of move (coefficient: 0.127, p-value<0.01 in column 6 of Table 3) and turn insignificant in the subsequent years. The results in columns 9 to 12 of Table 3 show that after the move, the fraction of new collaborators to the total distinct co-authors for mobile scientists is slightly larger than that for their statistic matched peers by 3.3% (coefficient: 0.033, p-value<0.01), and accordingly the fraction of previous collaborators for mobile scientists is significantly smaller than that for unmoved matched scientists by 3.3% (coefficient: -0.033, p-value<0.01). The above evidence shows that despite the positive effect of mobility on the absolute number of new collaborators and previous collaborators, moved scientists are more likely to collaborate with co-authors they have not collaborated with in the past after they move to a new affiliation, compared to their unmoved peers.

After mobility, upward movers, and non-upward movers have 17.0% (coefficient: 0.170, p<0.01 in column 7 of Table 4) and 14.7% (coefficient: 0.147, p<0.01 in column 7 of Table 5) more new collaborators than their matched unmoved peers, respectively. Despite the positive impact of upward mobility on the number of previous collaborators (coefficient:0.053, p-value<0.01 in column 5 of Table 4), the magnitude of the positive effect is small and is only significant in the move year. Furthermore, the fraction of new collaborator for upward movers is significantly larger than that for unmoved scientists by 2.9% (coefficient: 0.029, p-value<0.01 in column 11 of Table 4). Similar positive effect is found for non-upward move, since the fraction of new collaborator for non-upward movers is significantly larger than that for unmoved scientists by 2.7% (coefficient: 0.027, p-value<0.01 in column 11 of Table 5). The finding suggests that after moving to a new affiliation, regardless of the direction of the move, scientists have a higher probability to construct new collaboration.

To ensure our results regarding the main effect of mobility on productivity and collaboration are robust for scientists with a lower level of productivity, we conduct a robustness check by lowering the minimum of total publications of scientists to 5. The main effects of mobility on productivity and collaboration holds for 97,056 scientists who produced no fewer than 5 publications. 9.03% of those sampled scientists generated 12,404 mobility events in which upward mobility accounts for 33.38%. The PSM-DID estimates based on those scientists are shown in Table A9 to A11 in SI.

## 5. Discussion and conclusions

With the rapid globalization of science, mobility is considered an important driver of scientific progress (Wagner, Whetsell, Baas, & Jonkers, 2018). The tension between positive effects as a result of better match quality and knowledge spillover, and negative effects associated with disruption of routines and social capital make the effect of scientists' mobility on their productivity and collaboration remain to be further investigated.

In this study, we show a positive effect of domestic mobility on research quantity of scientists, regardless of the direction of their move. In contrast with the existing literature that documents a short-term negative effect (Fernández-Zubieta et al., 2015), we do not observe an initial negative effect following mobility, even for non-upward moved scientists. It may imply that the benefits resulting from mobility outweigh its costs, and improve scientists' research quantity after a job change. Generally, we do not observe that mobility increases publications' citations. This finding is consistent with the view that it is not the institution that creates the impact; it is the individual researchers that make an institution (Fortunato et al., 2018). Scientific impacts depend on the reception of the work by the scientific community and are less determined by institutions' reputations. Our sampled scientists are relatively productive with a total productivity of no fewer than 10 papers. For prolific scientists, institutions' reputation is less of a crucial driver for their scientific success than scientists' own research ability, diligence and relevance (Deville et al., 2014; Fortunato et al., 2018).

Our results show that scientists experienced a gain in the number of publications after moving to a new affiliation, regardless of the direction of the move. A higher-tier university might provide a better research environment or richer scientific resources, while other benefits caused by moving to a lower-tier or a same-tier university could also facilitate scientific productivity, such as a better match quality or higher level of relevance (Fortunato et al., 2018). It appears that the tier or rank of mobility destination may not matter as both upward and non-upward moves increase the number of publications and expand scientific ties. When making a decision to move, the reputation of the destination affiliation should not be the only factor scientists take into account and it is more important to consider the match quality and other benefits scientists could acquire after the move.

It is noticed that there is an immediate increase in the number of publications for moved scientists in the year of the move. Hiring new faculty, especially researchers from high-tier universities, might be a short-term strategy for departments to improve their productivity and visibility (Fernández-Zubieta et al., 2015), which might prompt moved scientists to show their high productivity immediately after they join the new affiliations. We further notice that in the second and third years after move, the increase in the number of publications becomes lower relative to the year of the move. The results show an overall positive effect of mobility, while the slow-down in the second and third years after the move might still indicate adaption costs and potential administrative burdens that might hinder scientists' research performance.

We further find that after mobility, moved scientists obtain more new collaborators than static scientists, while scientists' collaboration with their previous collaborators is not hampered by mobility and slightly increases if scientists moved to a higher-tier university. This is evidenced by the slightly increased co-authorship with previous collaborators after the mobility. Subsequently, we observe that the fraction of co-authorship with previous collaborators to the total distinct collaborators decreases after mobility. In other words, while scientists keep working with previous collaborators, they are more inclined to construct new collaborative relationships after they move to a new affiliation. Mobility enables scientists to construct more new scientific ties as geographical proximity and co-location with new colleagues in the affiliations scientists moved to facilitate personal interaction and a face-to-face meeting, which is beneficial to establish the collaboration ties with their new colleagues. This may account for the increased share of new collaborators after scientists move. Besides, in contrast to a detrimental effect of transitioning to a new affiliation on scientists' social capital constructed in their former affiliations proposed by some researchers (Groysberg et al., 2008), we find that the collaboration ties scientists have already developed still remain after they move to a new affiliation and even slightly increase. This indicates that scientists' collaboration ties formed in their former affiliations are enduring and portable to the new affiliations they join. In addition, moving to a new affiliation creates a better research environment where scientists could become more productive, which might increase their demands to collaborate and thus we observe a slight increase in the number of previous collaborators in the year after the move.

Despite a selection criterion we use to construct the research sample as introduced above, this case study can be a good representation for the investigation of domestic scientific mobility, especially in the context of emerging economies for multiple reasons. Exploring Chinese scientists' domestic mobility offers a decent example in which the findings may be applied to the context of emerging economies. Furthermore, it would be more precise to capture scientists' mobility if those scientists incline to publish in English or the field they are working in is highly internationalized. Focusing on scientists in computer science, a discipline which is highly internationalized compared to most other disciplines in China (Fiala & Tutoky, 2017), facilitates enhancing the accuracy and generalizability of the findings to the disciplines where the objects of study are universal and publishing in English is preferred, e.g., chemistry, engineering and other natural sciences.

There are some limitations. Temporal or short-term mobility might not be captured in this study if there is no publication associated with scientists' move. Besides, although the PSM-DID estimator is effective in mitigating the endogeneity problem of the treatment based on observable variables and time-invariant unobservable variables, compared to randomized control trials, the PSM-DID estimator still has the limitation that unmeasured confounding variables varying over time might be present. The large-scale database used in this study helps improve the objectivity and generalizability of results compared to those on a smaller scale or based on self-reported data, which might alleviate this problem. Due to a trade-off between the precision of tracking scientists' mobility and the coverage of scientists across different levels of productivity, this study is conducted based on the sample of scientists with a total productivity of no fewer than 10. The findings of this study may not be applicable to less productive scientists. Although author disambiguation is automatically resolved in Scopus, it is not "perfectly accurate" as the Scopus Author ID could be mis-integrated or

mis-separated (Kawashima & Tomizawa, 2015). To get a more accurate measure of scientists' mobility, survey data and biographical records could be complementary data sources. The international mobility of Chinese scientists should be investigated as it could impact scientists' careers deeply and the findings of this study might not hold if scientists move across national borders. Non-voluntary influential factors of mobility, such as spouse hiring and other family factors, should be investigated further as these reasons might account for scientists' mobility to a same-tier university or downward mobility.

## CRediT authorship contribution statement

**Meijun Liu:** Conceptualization, Methodology, Data curation, Formal analysis, Writing – original draft. **Xiao Hu:** Conceptualization, Methodology, Writing – review & editing, Supervision.

## Acknowledgements

We thank National Bureau of Economic Research for its support in data preparation. This work is funded by the Science and Technology Commission of Shanghai Municipality (Science and Technology Innovation Action Plan of 2021-Soft Science Key Project, Grant Number: 21692196000; Shanghai Pujiang Program, Grant Number: 21PJC026). We also appreciate the anonymous reviewers whose suggestions helped improve this manuscript.

## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.joi.2022.101311.

## References

Ackers, L. (2008). Internationalisation, mobility and metrics: A new form of indirect discrimination? *Minerva, 46*(4), 411–435.
Agrawal, A., Cockburn, I., & McHale, J. (2006). Gone but not forgotten: knowledge flows, labor mobility, and enduring social relationships. *Journal of Economic Geography, 6*(5), 571–591.
Agrawal, A. K. (2013). The impact of investor protection law on corporate policy and performance: Evidence from the blue sky laws. *Journal of Financial Economics, 107*(2), 417–435.
Aksnes, D. W., Rørstad, K., Piro, F. N., & Sivertsen, G. (2013). Are mobile researchers more productive and cited than non-mobile researchers? A large-scale study of Norwegian scientists. *Research evaluation, 22*(4), 215–223.
Allen, R. M. (2017). A comparison of China's "Ivy League" to other peer groupings through global university rankings. *Journal of Studies in International Education, 21*(5), 395–411.
Almeida, P., Hohberger, J., & Parada, P. (2011). Individual scientific collaborations and firm-level innovation. *Industrial and Corporate Change, 20*(6), 1571–1599.
Aman, V. (2018). Does the Scopus author ID suffice to track scientific international mobility? A case study based on Leibniz laureates. *Scientometrics, 117*(2), 705–720.
Audretsch, D. B., & Stephan, P. E. (1996). Company-scientist locational links: The case of biotechnology. *The American Economic Review, 86*(3), 641–652.
Azoulay, P., Zivin, J. S. G., & Sampat, B. N. (2011). The diffusion of scientific knowledge across time and space. *Evidence from professional transitions for the superstars of medicine* (0898-2937)Retrieved from.
Azoulay, P., Ganguli, I., & Zivin, J. G. (2017). The mobility of elite life scientists: Professional and personal determinants. *Research Policy, 46*(3), 573–590.
Bäker, A. (2015). Non-tenured post-doctoral researchers' job mobility and research output: An analysis of the role of research discipline, department size, and coauthors. *Research Policy, 44*(3), 634–650.
Bhugra, D. (2003). Migration and depression. *Acta Psychiatrica Scandinavica, 108*(s418), 67–72. 10.1034/j.1600-0447.108.s418.14.x.
Blundell, R., & Costa Dias, M. (2000). Evaluation methods for non-experimental data. *Fiscal Studies, 21*(4), 427–468.
Blundell, R., & Dias, M. C. (2009). Alternative approaches to evaluation in empirical microeconomics. *Journal of Human Resources, 44*(3), 565–640.
Borjas, G. J., & Doran, K. B. (2012). The collapse of the Soviet Union and the productivity of American mathematicians. *The Quarterly Journal of Economics, 127*(3), 1143–1203.
Bozeman, B., & Boardman, C. (2014). *Research collaboration and team science: A state-of-the-art review and agenda*. Springer.
Brzezinski, M. (2015). Power laws in citation distributions: evidence from Scopus. *Scientometrics, 103*(1), 213–228. 10.1007/s11192-014-1524-z.
Bu, Y., Li, H., Wei, C., Liu, M., & Li, J. (2020). On the relationship between supervisor–supervisee gender difference and scientific impact of doctoral dissertations: Evidence from Humanities and Social Sciences in China. *Journal of Information Science*. 10.1177/0165551520969935.
Cañibano, C., Otamendi, J., & Andújar, I. (2008). Measuring and assessing researcher mobility from CV analysis: the case of the Ramón y Cajal programme in Spain. *Research evaluation, 17*(1), 17–31.
Cañibano, C., D'Este, P., Otamendi, F. J., & Woolley, R. (2020). Scientific careers and the mobility of European researchers: an analysis of international mobility by career stage. *Higher education, 80*(6), 1175–1193.
Cadot, O., Fernandes, A. M., Gourdon, J., & Mattoo, A. (2015). Are the benefits of export support durable? Evidence from Tunisia. *Journal of International Economics, 97*(2), 310–324.
Davies, S., & Zarifa, D. (2012). The stratification of universities: Structural inequality in Canada and the United States. *Research in Social Stratification and Mobility, 30*(2), 143–158.
De Filippo, D., Casado, E. S., & Gómez, I. (2009). Quantitative and qualitative approaches to the study of mobility and scientific performance: a case study of a Spanish university. *Research evaluation, 18*(3), 191–200.
Dehejia, R. (2005). Practical propensity score matching: a reply to Smith and Todd. *Journal of Econometrics, 125*(1-2), 355–364.
Deville, P., Wang, D., Sinatra, R., Song, C., Blondel, V. D., & Barabási, A.-L. (2014). Career on the move: Geography, stratification, and scientific impact. *Scientific reports, 4*, 4770.
Dokko, G., & Rosenkopf, L. (2010). Social capital for hire? Mobility of technical professionals and firm influence in wireless standards committees. *Organization Science, 21*(3), 677–695.
Fangmeng, T. (2016). Brain circulation, diaspora and scientific progress: A study of the international migration of Chinese scientists, 1998–2006. *Asian and Pacific migration journal, 25*(3), 296–319.
Fernández-Zubieta, A., Geuna, A., & Lawson, C. (2015). Mobility and productivity of research scientists. In *Global mobility of research scientists* (pp. 105–131). Elsevier.
Fernández-Zubieta, A., Geuna, A., & Lawson, C. (2016). Productivity pay-offs from academic mobility: should I stay or should I go? *Industrial and Corporate Change, 25*(1), 91–114.
Fernandez-Zubieta, A., Geuna, A., & Lawson, C. (2013). Researchers' mobility and its impact on scientific productivity.
Fiala, D., & Tutoky, G. (2017). Computer science papers in Web of Science: A bibliometric analysis. *Publications, 5*(4), 23.
Fleming, L. (2001). Recombinant uncertainty in technological search. *Management Science, 47*(1), 117–132.

Fontes, M., Videira, P., & Calapez, T. (2013). The impact of long-term scientific mobility on the creation of persistent knowledge networks. *Mobilities, 8*(3), 440–465. 10.1080/17450101.2012.655976.
Fortunato, S., Bergstrom, C. T., Börner, K., Evans, J. A., Helbing, D., Milojević, S., & Uzzi, B. (2018). Science of science. *Science, 359*(6379), eaao0185.
Franzoni, C., Scellato, G., & Stephan, P. (2014). The mover's advantage: The superior performance of migrant scientists. *Economics Letters, 122*(1), 89–93.
Frenken, K., Hardeman, S., & Hoekman, J. (2009). Spatial scientometrics: Towards a cumulative research program. *Journal of Informetrics, 3*(3), 222–232.
Furukawa, T., Shirakawa, N., & Okuwada, K. (2011). Quantitative analysis of collaborative and mobility networks. *Scientometrics, 87*(3), 451–466.
Ganguli, I. (2015). Who leaves and who stays? Evidence on immigrant selection from the collapse of Soviet science. In *Global Mobility of Research Scientists* (pp. 133–154). Elsevier.
Gaughan, M., & Robin, S. (2004). National science training policy and early scientific careers in France and the United States. *Research policy, 33*(4), 569–581.
Gibson, J., & McKenzie, D. (2012). The economic consequences of 'brain drain'of the best and brightest: Microeconomic evidence from five countries. *The Economic Journal, 122*(560), 339–375.
Gill, S. (2010). The homecoming: an investigation into the effect that studying overseas had on Chinese postgraduates' life and work on their return to China. *Compare: A Journal of Comparative and International Education, 40*(3), 359–376. 10.1080/03057920903464555.
Girma, S., & Görg, H. (2007). Evaluating the foreign ownership wage premium using a difference-in-differences matching approach. *Journal of International Economics, 72*(1), 97–112.
Grogger, J., & Hanson, G. H. (2011). Income maximization and the selection and sorting of international migrants. *Journal of Development Economics, 95*(1), 42–57.
Groysberg, B., Lee, L.-E., & Nanda, A. (2008). Can they take it with them? The portability of star knowledge workers' performance. *Management Science, 54*(7), 1213–1230.
Hargadon, A., & Sutton, R. I. (1997). Technology brokering and innovation in a product development firm. *Administrative science quarterly*, 716–749.
Heckman, J. J., Ichimura, H., & Todd, P. E. (1997). Matching as an econometric evaluation estimator: Evidence from evaluating a job training programme. *The Review of Economic Studies, 64*(4), 605–654.
Hoisl, K. (2007). Tracing mobile inventors—the causality between inventor mobility and inventor productivity. *Research Policy, 36*(5), 619–636.
Jaffe, A. B., Trajtenberg, M., & Henderson, R. (1993). Geographic localization of knowledge spillovers as evidenced by patent citations. *The Quarterly Journal of Economics, 108*(3), 577–598.
Jones, B. F., & Weinberg, B. A. (2011). Age dynamics in scientific creativity. *Proceedings of the National Academy of Sciences, 108*(47), 18910–18914.
Jones, K. (2021). Brokered discrimination for a fee: the incompatibility of domestic work placement agencies with rights-based global governance of migration. *Global Public Policy and Governance, 1*, 300–320. 10.1007/s43508-021-00017-8.
Jonkers, K., & Cruz-Castro, L. (2013). Research upon return: The effect of international mobility on scientific ties, production and impact. *Research Policy, 42*(8), 1366–1377.
Jonkers, K., & Tijssen, R. (2008). Chinese researchers returning home: Impacts of international mobility on research collaboration and scientific productivity. *Scientometrics, 77*(2), 309–333.
Katz, J. S. (1994). Geographical proximity and scientific collaboration. *Scientometrics, 31*(1), 31–43.
Kawashima, H., & Tomizawa, H. (2015). Accuracy evaluation of Scopus Author ID based on the largest funding database in Japan. *Scientometrics, 103*(3), 1061–1071.
Lariviere, V., Sugimoto, C. R., Tsou, A., & Gingras, Y. (2014). Team size matters: Collaboration and scientific impact since 1900: on the relationship between collaboration and scientific impact since 1900. *Journal of the Association for Information Science & Technology, 66*(7).
Lehmann, S., Lautrup, B., & Jackson, A. D. (2003). Citation networks in high energy physics. *Physical review E, 68*(2), Article 026113. 10.1103/PhysRevE.68.02611.
Lenzi, C. (2009). Patterns and determinants of skilled workers' mobility: evidence from a survey of Italian inventors. *Economics of Innovation and New Technology, 18*(2), 161–179.
Levin, S. G., & Stephan, P. E. (1991). Research productivity over the life cycle: Evidence for academic scientists. *The American Economic Review*, 114–132.
Li, F., & Tang, L. (2019). When international mobility meets local connections: Evidence from China. *Science and Public Policy, 46*(4), 518–529.
Liu, M., & Hu, X. (2021). Will collaborators make scientists move? A generalized propensity score analysis. *Journal of Informetrics, 15*(1), Article 101113. 10.1016/j.joi.2020.101113.
Liu, M., Hu, X., Wang, Y., & Shi, D. (2018). Survive or perish: Investigating the life cycle of academic journals from 1950 to 2013 using survival analysis methods. *Journal of Informetrics, 12*(1), 344–364.
Liu, M., Bu, Y., Chen, C., Xu, J., Li, D., Leng, Y., & Sung, M. (2021). Pandemics are catalysts of scientific novelty: Evidence from COVID-19. *Journal of the Association for Information Science and Technology*.
Liu, M., Zhang, N., Hu, X., Jaiswal, A., Xu, J., Chen, H., & Bu, Y. (2022). Further Divided Gender Gaps in Research Productivity and Collaboration during the COVID-19 Pandemic: Evidence from Coronavirus-related Literature. *Journal of Informetrics*, Article 101295.
Liu, M., Shi, D., & Li, J. (2017). Double-edged sword of interdisciplinary knowledge flow from hard sciences to humanities and social sciences: Evidence from China. *PLoS ONE, 12*. 10.1371/journal.pone.0184977.
Long, J. S., Allison, P. D., & McGinnis, R. (1993). Rank advancement in academic careers: Sex differences and the effects of productivity. *American Sociological Review*, 703–722.
Manuel, T., Shiff, G., & Melamed, R. (2006). *The "Names Game": Harnessing Inventors' Patent Data for Economic Research*. Working Paper Series. National Bureau of Economic Research. Retrieved from http://www.nber.org/papers/w12479
Martin-Rovet, D. (2003). *Opportunities for outstanding young scientists in Europe to create an independent research team*. Strasbourg: European Science Foundation.
Melkers, J., & Kiopa, A. (2010). The social capital of global ties in science: The added value of international collaboration. *Review of Policy Research, 27*(4), 389–414.
Merton, R. K. (1968). The Matthew effect in science: The reward and communication systems of science are considered. *Science, 159*(3810), 56–63.
Moser, P., Voena, A., & Waldinger, F. (2014). German Jewish émigrés and US invention. *American Economic Review, 104*(10), 3222–3255.
Netz, N., Hampel, S., & Aman, V. (2020). What effects does international mobility have on scientists' careers? A systematic review. *Research Evaluation, 29*(3), 327–351.
Ponds, R., Van Oort, F., & Frenken, K. (2007). The geographical and institutional proximity of research collaboration. *Papers in regional science, 86*(3), 423–443.
Robinson-Garcia, N., Sugimoto, C. R., Murray, D., Yegrosyegros, A., Lariviere, V., & Costas, R. (2019). The many faces of mobility: Using bibliometric data to measure the movement of scientists. *Journal of Informetrics, 13*(1), 50–63.
Rosenbaum, P. R., & Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika, 70*(1), 41–55.
Salmi, J. (2010). *The challenge of establishing world-class research universities in developing countries* (p. 271). University Research for Innovation.
Scellato, G., Franzoni, C., & Stephan, P. (2012). Mobile scientists and international networks (No. w18613). *National Bureau of Economic Research*.
Scellato, G., Franzoni, C., & Stephan, P. (2015). Migrant scientists and international networks. *Research Policy, 44*(1), 108–120.
Shin, J. C., & Jung, J. (2014). Academics job satisfaction and job stress across countries in the changing academic environments. *Higher education, 67*(5), 603–620. 10.1007/s10734-013-9668-y.
Shin, J. C., Arimoto, A., Cummings, W. K., & Teichler, U. (2014). *Teaching and Research in Contemporary Higher Education: An Overview*. Netherlands: Springer.
Singh, J., & Agrawal, A. (2011). Recruiting for ideas: How firms exploit the prior inventions of new hires. *Management Science, 57*(1), 129–150.
Smith, J. A., & Todd, P. E. (2005). Does matching overcome LaLonde's critique of nonexperimental estimators? *Journal of Econometrics, 125*(1-2), 305–353.
Somaya, D., Williamson, I. O., & Lorinkova, N. (2008). Gone but not lost: The different performance impacts of employee mobility between cooperators versus competitors. *Academy of Management Journal, 51*(5), 936–953.
Sugimoto, C. R., Robinson-García, N., Murray, D. S., Yegros-Yegros, A., Costas, R., & Larivière, V. (2017). Scientists have most impact when they're free to move. *Nature News, 550*(7674), 29.
Tartari, V., Di Lorenzo, F., & Campbell, B. A. (2018). Another roof, another proof": the impact of mobility on individual productivity in science. *The Journal of Technology Transfer*, 1–28.
Topel, R. H., & Ward, M. P. (1992). Job mobility and the careers of young men. *The Quarterly Journal of Economics, 107*(2), 439–479.

Van Heeringen, A., & Dijkwel, P. (1987). The relationships between age, mobility and scientific productivity. Part I: Effect of mobility on productivity. *Scientometrics, 11*(5-6), 267–280.

Velema, T. A. (2012). The contingent nature of brain gain and brain circulation: their foreign context and the impact of return scientists on the scientific community in their country of origin. *Scientometrics, 93*(3), 893–913.

Wagner, C. S., Whetsell, T., Baas, J., & Jonkers, K. (2018). Openness and impact of leading scientific countries. *Frontiers in Research Metrics and Analytics, 3*, 10.

Wagner, C. S., Whetsell, T. A., & Mukherjee, S. (2019). International research collaboration: Novelty, conventionality, and atypicality in knowledge recombination. *Research policy, 48*(5), 1260–1270.

Wagner, C. (2008). The New Invisible College: Science for development (Washington. DC: Brookings.

Waldinger, F. (2012). Peer effects in science: Evidence from the dismissal of scientists in Nazi Germany. *The Review of Economic Studies, 79*(2), 838–861.

Wang, J., Hooi, R., Li, A. X., & Chou, M.-h. (2019). Collaboration patterns of mobile academics: The impact of international mobility. *Science and Public Policy, 46*(3), 450–462.

Wang, Y., Hu, R., & Liu, M. (2017). The Geotemporal Demographics of Academic Journals from1950 to 2013 according to Ulrich's Database. *Journal of Informetrics, 11*, 655–671. 10.1016/j.joi.2017.05.006.

Woods, D. (2022). Populism has relationship issues: collider effects and immigration policy in Italy. *Global Public Policy and Governance, 2*, 5–21. 10.1007/s43508-022-00035-0.

Xu, H., Bu, Y., Liu, M., Chen, W., Sun, M., Zhang, Y., … Ding, Y. (2022). Team power dynamics and team impact: New perspectives on scientific collaboration using career age as a proxy for team power. *Journal of the Association for Information Science and Technology*. 10.1002/asi.24653.

Zong, X., & Zhang, W. (2019). Establishing world-class universities in China: deploying a quasi-experimental design to evaluate the net effects of Project 985. *Studies in Higher Education, 44*(3), 417–431.

Zucker, L. G., Darby, M. R., & Torero, M. (2002). Labor mobility from academe to commerce. *Journal of Labor Economics, 20*(3), 629–660.